INJECTIVITY OF THE MAP OF QUOTIENT SHEAVES UNDER RESTRICTION OF A GROUPOID SCHEME

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Allowed standard prerequisites (editorial).

- Lemmas lemma-groupoid-pre-equivalence (0232) and lemma-quotient-pre-equivalence-relation-restrict (045Z). cited at proof lines 4107-4109. Injectivity of U'/R' -> U/R.
- Section sites-section-sheaves-injective (00WL). cited at proof lines 4109-4112. Characterization of surjective maps of sheaves.
- Equation equation-quotient-presheaf (02VF). cited at proof lines 4120-4124. The presheaf quotient equivalence relation.
- Lemma sites-lemma-covering-surjective-after-sheafification (00WT). cited at proof lines 4130-4132. Coverings induce sheaf surjections.

Source reference. Original lemma 0232: sources/stacks-chow/tags.txt, source lines 2727–2727. Cited at proof lines 4107-4109 (combined with 045Z) for injectivity; load-bearing.

Source reference. Original lemma 045Z: sources/stacks-chow/tags.txt, source lines 5454–5454. Cited at proof lines 4107-4109 (combined with 0232) for injectivity; load-bearing.

Source reference. Original section 00WL: sources/stacks-chow/tags.txt, source lines 1187–1187. Cited at proof lines 4109-4112 for the characterization of surjectivity used; navigation for the criterion.

Source reference. Original equation 02VF: sources/stacks-chow/tags.txt, source lines 3766–3766. Cited at proof lines 4120-4124 for the equivalence notation and the witnessing element pr_1 of f_i; load-bearing.

Source reference. Original lemma 00WT: sources/stacks-chow/tags.txt, source lines 1195–1195. Cited at proof lines 4130-4132 to justify the first sufficient condition; load-bearing for the final paragraph.

Lemma 0.2. *Let $\tau \in \{Zariski, étale, fppf, smooth, syntomic\}$. Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Let $g : U' \to U$ a morphism of*

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schemes over S. Let (U', R', s', t', c') be the restriction of (U, R, s, t, c) to U'. Assume U, U', R, S are objects of a τ-site Sch_τ. The map of quotient sheaves

$$U'/R' \longrightarrow U/R$$

is injective. If the composition

$$U' \times_{g,U,t} R \xrightarrow[pr_1]{} R \xrightarrow[s]{} U, \quad \text{with } h \text{ the composite}$$

defines a surjection of sheaves in the τ-topology then the map is bijective. This holds for example if $\{h : U' \times_{g,U,t} R \to U\}$ is a τ-covering, or if $U' \to U$ defines a surjection of sheaves in the τ-topology, or if $\{g : U' \to U\}$ is a covering in the τ-topology.

Proof. Injectivity follows on combining Lemmas 0232 and 045Z. To see surjectivity (see Sites, Section 00WL for a characterization of surjective maps of sheaves) we argue as follows. Suppose that T is a scheme and $\sigma \in U/R(T)$. There exists a covering $\{T_i \to T\}$ such that $\sigma|_{T_i}$ is the image of some element $f_i \in U(T_i)$. Hence we may assume that σ is the image of $f \in U(T)$. By the assumption that h is a surjection of sheaves, we can find a τ-covering $\{\varphi_i : T_i \to T\}$ and morphisms $f_i : T_i \to U' \times_{g,U,t} R$ such that $f \circ \varphi_i = h \circ f_i$. Denote $f'_i = \mathrm{pr}_0 \circ f_i : T_i \to U'$. Then we see that $f'_i \in U'(T_i)$ maps to $g \circ f'_i \in U(T_i)$ and that $g \circ f'_i \sim_{T_i} h \circ f_i = f \circ \varphi_i$ notation as in (02VF). Namely, the element of $R(T_i)$ giving the relation is $\mathrm{pr}_1 \circ f_i$. This means that the restriction of σ to T_i is in the image of $U'/R'(T_i) \to U/R(T_i)$ as desired.

If $\{h\}$ is a τ-covering, then it induces a surjection of sheaves, see Sites, Lemma 00WT. If $U' \to U$ is surjective, then also h is surjective as s has a section (namely the neutral element e of the groupoid scheme). □